

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

ANS ::

1. **CMOS Battery Failure**

**Issue:** The CMOS battery may be weak or dead.

2. **Keyboard Error or No Keyboard Detected**

**Issue:** Keyboard may not be connected properly or may be faulty.

3. **Memory Error / RAM Error**

**Issue:** RAM may be loose, damaged, or incompatible.

4. **No Boot Device Found**

**Issue:** HDD/SSD may not be detected or the boot settings may be incorrect.

5. **CPU Fan Error**

**Issue:** CPU fan may not be connected or may not be working properly.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

Ans ::

When we turn on a computer, the **BIOS checks all the hardware**. This checking process is called **POST**.

If the computer makes **3 short beeps** again and again and **nothing appears on the screen**, it usually means there is a **problem with the RAM**.

The computer cannot properly detect or check the RAM. Because of this, the computer **cannot start properly**, and the monitor may show **no display**.

### **Troubleshooting Steps:**

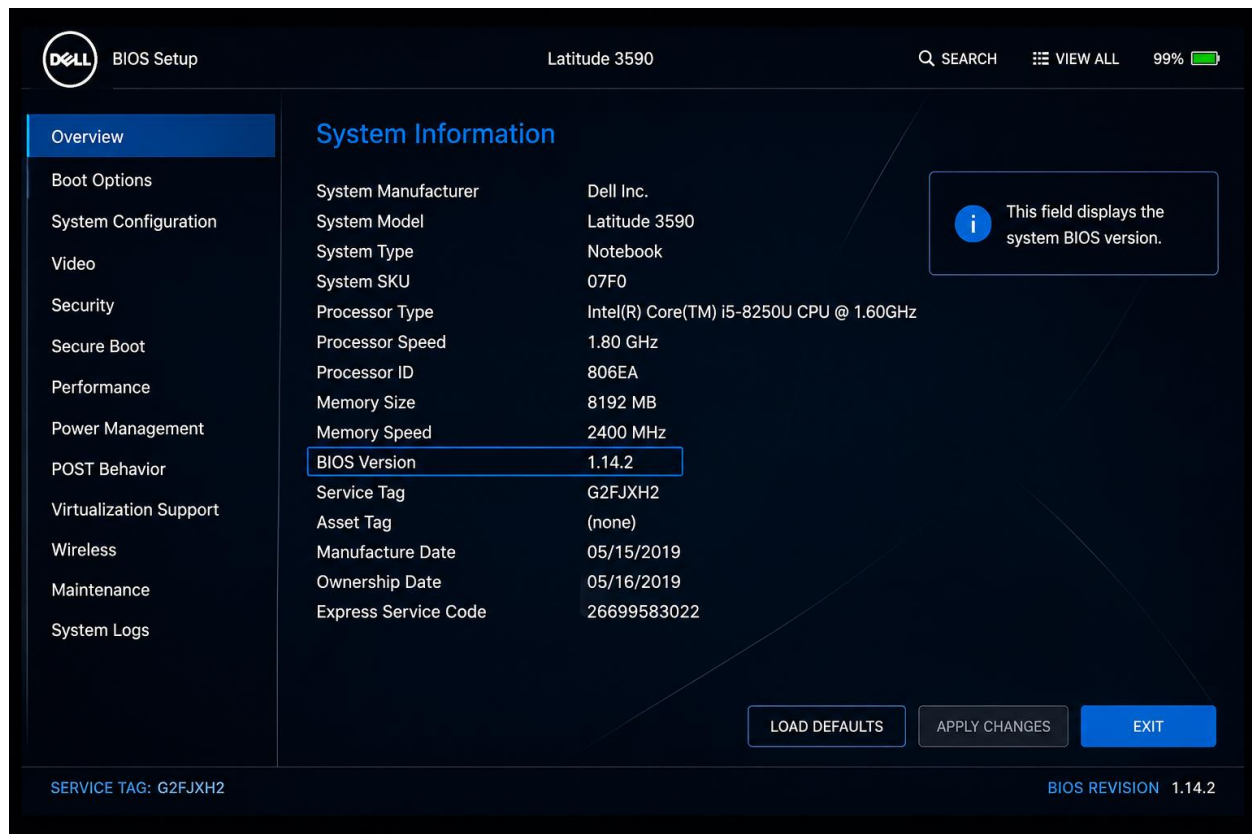
#### **1. Remove and Reinstall the RAM:**

First, turn off the computer and unplug the power cable. Remove the RAM from its slot and then carefully install it again. Make sure that the RAM is properly and firmly fitted into the slot.

#### **2. Test the RAM One by One:**

If the computer has two or more RAM sticks, remove one RAM stick and try starting the computer with only one stick. Then test the other RAM stick separately. This can help identify if one of the RAM sticks is faulty.

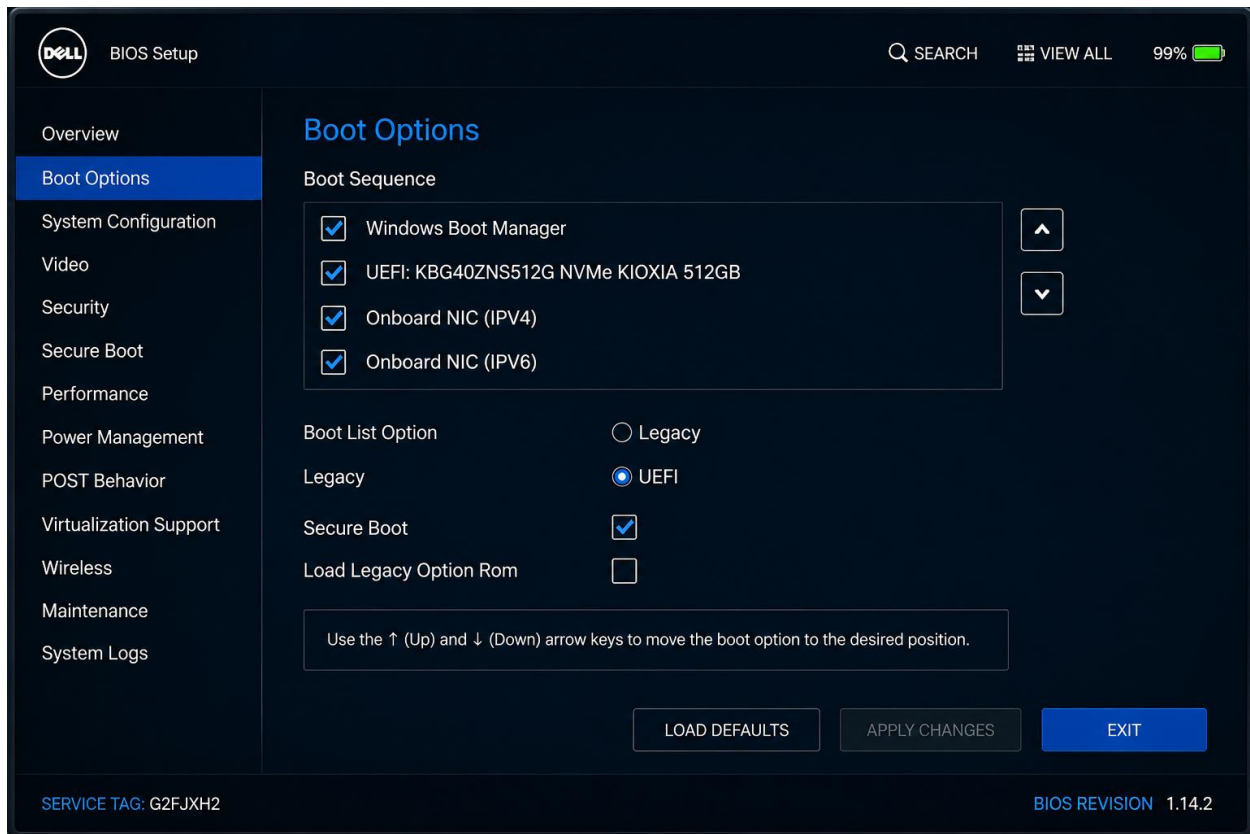
3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



➤ This is the **Dell BIOS Setup screen**. It shows important information about the laptop.

- **Manufacturer:** Dell Inc.
- **Model:** Latitude 3590
- **BIOS Version:** 1.14.2
- **Memory (RAM):** 8192 MB (8 GB)

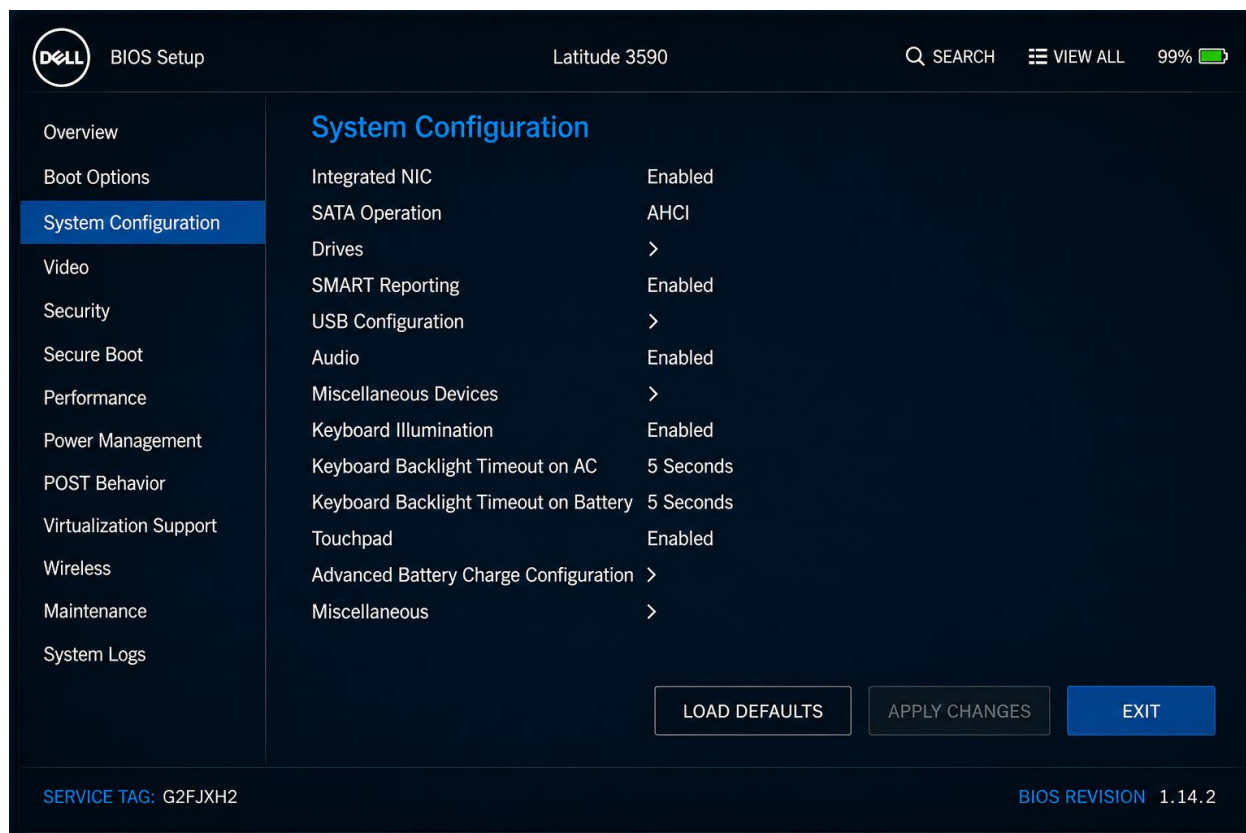
➤ **In simple words:** This BIOS screen helps us check the computer's hardware and BIOS information.



➤ This image shows the **Dell BIOS Setup – Boot Options** screen.

- **Windows Boot Manager** is selected as the first boot option.
- **UEFI** mode is enabled.
- **Secure Boot** is enabled.
- It also shows the **BIOS Revision: 1.14.2**.

➤ **In simple words:** This screen controls **how the computer starts (boots)** and which device it uses to start Windows.



➤ I accessed the **BIOS Setup Utility** and checked the main BIOS screen.

- **Motherboard Manufacturer:** Dell Inc.
- **BIOS Version:** 1.14.2

➤ **Screenshot:** The BIOS main screen is attached as evidence.

4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.<br><br><em><strong>Hint:</strong> Think about the CMOS battery and BIOS settings reset.</em>

## CMOS CHECKSUM ERROR – DEFAULTS LOADED

### WHAT DOES IT MEAN?

 This error means the BIOS settings stored in **CMOS memory** are lost, corrupted, or reset. Because of this, the system has loaded **default BIOS settings** to start.

### WHY DOES IT HAPPEN?

 It usually happens when the CMOS battery is weak or dead, or when BIOS settings get corrupted.

### HOW TO RESOLVE IT?

- **Power Off & Unplug**  
Turn off the system and unplug the power cable.
- **Replace CMOS Battery**  
Open the case and replace the old CMOS battery with a new compatible battery.
- **Enter BIOS Setup**  
Power on the system and press the required key (Del/F2/F10) to enter BIOS.
- **Set BIOS Settings**  
Set the correct Date, Time and other required settings. Save and Exit.
- **Restart System**  
Restart the system and check if the error appears again.

### IN SIMPLE WORDS:

 CMOS checksum error means BIOS settings were lost or reset, usually because of a weak or dead CMOS battery. Replace the battery and set BIOS settings again to fix the issue.



“**CMOS checksum error – Defaults loaded**” means that the computer has detected a problem with the **CMOS/BIOS settings**. The stored BIOS settings may have been lost, corrupted, or reset.

Usually, this happens because the **CMOS battery is weak or dead**. As a result, the BIOS loads the **default settings** when the computer starts.

### **How to Resolve It:**

1. **Replace the CMOS battery:** Turn off the computer, unplug it, and replace the old CMOS battery with a new compatible battery.
  2. **Set BIOS settings again:** Enter the BIOS Setup, check the date and time and other required settings, then **Save and Exit**.
  3. Restart the computer and check whether the error appears again.
- **CMOS error = BIOS settings were lost/reset, usually because of a weak CMOS battery.**